M-CORE: A Novel Approach for Land Cover Fraction Mapping Using Multisite Spectral Libraries

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***Abstract*— The concept of a generic spectral library (GSL), i.e., a structured collection of image spectra sampled from various optical remote sensing images covering different sites and points in time, has been suggested in previous studies to facilitate the tedious process of producing training data for remote sensing-based land cover (LC) mapping. When using multisource libraries collected over different sites, library pruning approaches are needed to extract an apt set of labeled spectra to perform mapping on a specific area. Library pruning mainly aims to discard image-irrelevant and redundant spectra, while limiting spectral confusion during the mapping process. Most library pruning approaches focus only on one of these aspects, which has been shown to negatively affect mapping accuracies. This article emphasizes the need for a multistep approach to optimize a GSL for site-specific mapping. We propose a new library pruning method called M-CORE, specifically designed to facilitate LC fraction mapping. The method extends multiple signal classification (MUSIC) with a confusion reduction (CORE) component. Vegetation-impervious-soil (VIS) fraction mapping experiments on a Sentinel-2 image of Brussels, using a GSL with and without local spectra, show the added value of M-CORE over previously proposed library pruning methods and demonstrate the feasibility of GSL-based mapping in an urban context.**

***Index Terms*— Generic spectral library (GSL), land cover (LC), library pruning, M-CORE, spectral libraries, spectral unmixing, urban, vegetation-impervious-soil (VIS).**

I. Introduction

LAND cover mapping requires the availability of reference data that link spectral responses to different land cover (LC) types. Pure spectra representing each LC type are referred to as endmembers (EMs). EM spectra are typically stored in spectral libraries. Spectral libraries used in remote sensing nowadays often contain spectral data from multiple images and/or sensors, and they are becoming larger and more diverse [1]–[4]. Dedicated data mining and learning approaches are becoming increasingly important to properly exploit them [5]. Unsurprisingly, spectral libraries are now often integrated in database management systems, which allows efficient querying and retrieval [1], [6], [7]. Examples of spectral library (collections) commonly used in the remote sensing community are Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) [1], Humboldt-Universität zu Berlin (HUB) [8], DLR [9], ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS) [4], Spectral Library of impervious Urban Material (SLUM) [10], Karlsruhe Library of Urban Material (KLUM) [11], and UCSB [3].

The evolution toward bigger and more diverse spectral libraries brings up the question to what extent spectral libraries used for training of image classifiers can be transferred across different sites, sensors, and scales while still yielding accurate mapping results [12]. This topic has been the subject of numerous studies [13]–[22]. Okujeni *et al.* [22] demonstrated that a generalized urban spectral library—containing, in their case, data from Berlin (Germany), Brussels (Belgium), and Santa Barbara (U.S.)—could enhance site transferability. However, accurate transferal of soil mapping spectra remained a challenge. Foody *et al.* [15] combined data from multiple sites in a generalized spectral library to train a neural network that predicts vegetation indices of forests. The results showed modest increases in site transferability. Furthermore, adding test site location as a dummy in the input layer of the neural network increased accuracy significantly. Schneider *et al.* [19], in a global urban mapping effort, managed to overcome the large differences between cities in different regions by stratification. The "urban ecoregions" they defined differed in climate, vegetation, urban morphology, population size, and the level of economic development. Training data used to classify an urban area were sampled only from that specific ecoregion.

For repeated mapping of multiple cities, the use of a generic spectral library (GSL), capturing the vast spectral variability present within different urban areas, would constitute a more economical alternative to the tedious development of image-specific libraries prior to mapping [12]. Large spectral libraries often contain vast amounts of spectra, typically much more than what is needed for mapping a specific site. Library pruning retrieves the relevant EM spectra from a larger library, reducing the computational load in LC mapping. Moreover, library pruning may also substantially increase mapping accuracy [12]. There are four types of pruning approaches that have been proposed in the literature: distance-based, estimator-based, image-based, and local approaches.

Distance-based pruning produces a spectral library subset by removing library entries that are spectrally alike, thus reducing redundancy of spectra. An example of distance-based pruning is an approach proposed by Dennison and Roberts [23] that uses endmember average RMSE (EAR) to evaluate the ability of each EM to model the spectra within its own class.

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EAR is calculated for each EM by averaging the root-mean-squared error (RMSE) of the set of models that use that EM to unmix all other spectra belonging to the same LC class. Minimum average spectral angle (MASA) is a similar selection metric [24], but it uses spectral angle instead of RMSE [25]. This metric is not affected by differences in illumination. Iterative spectral distancing (ISD) is originally an EM extraction technique but can also be used in a pruning context [26]. In contrast to EAR and MASA, which iteratively compare each spectrum with all other spectra, ISD evaluates each spectrum against the average of all spectra. Seeing that the number of unique pairs scales quadratically with library size, EAR and MASA may become computationally infeasible for larger libraries. ISD needs less calculations and can furthermore be used with any spectral similarity measure.

Estimator-based pruning produces a spectral library subset by identifying and removing library entries that can be modeled by a combination of other library spectra, using a specific estimator. An example of estimator-based pruning is iterative endmember selection (IES, [24]). This method retains an optimal library subset that produces the highest kappa value [27] when estimating all library spectra labels with multiple endmember spectral mixture analysis (MESMA) classification. IES works in two steps. First, the algorithm compares all possible pairs of EMs to find the two EMs that result in the highest kappa value for classifying the training library using a predefined classification method. Then, spectra are iteratively added and discarded from the training set until kappa converges to a maximum value. A disadvantage is that IES requires heavy computations due to its exhaustive sampling design. A less computationally heavy alternative is to use a genetic algorithm variant of IES [28].

Image-based pruning directly compares candidate EMs with spectra in the image to be classified [12]. Some image-based techniques select relevant EMs by individually comparing image pixels with library spectra based on distance metrics [29] or sparse unmixing [30]. However, these techniques are computationally demanding due to their pixelwise approach. Multiple signal classification (MUSIC) is an example of an image-based pruning algorithm [31] that operates on the image as a whole and as such is computationally less demanding. The technique calculates the Euclidean distance between library spectra and the subspace formed by the first eigenvectors of the covariance matrix of an image [31]. The user chooses a percentage-retained threshold which corresponds with a MUSIC distance below which spectra will be retained [31]. In the original approach, the number of eigenvectors is determined by an algorithm, called hyperspectral signal subspace identification by minimum error (HySime, [32]), which is based on signal noise decomposition. MUSIC has been shown to produce more accurate mapping results compared with other pruning approaches [12], [31].

Some techniques combine different pruning approaches and are referred to as hybrid. An example of this is automated MUSIC and spectral separability-based endmember selection (AMUSES), which refines the MUSIC technique by combining distance- and image-based pruning methods by selecting EM subsets with respect to the image to be processed, while accounting for internal redundancy [12]. The method starts by applying brightness normalization on the original image [33]. This is done, because the original MUSIC algorithm has been demonstrated to have a bias toward selecting bright spectra [34]. Then, spectra are sorted according to their MUSIC distance. A fraction of the lower distance spectra is then retained, while a fraction of the higher distance spectra is discarded. The rest of the spectra are evaluated so that only spectra that are sufficiently dissimilar from already selected spectra are retained [12].

Local pruning uses *a priori* classification results to subdivide the image in different regions and to subsequently steer the pruning process locally based on the classes that are present [35]. Furthermore, local pruning aims to improve mapping by discarding non-local spectra and as such decrease spectral confusion during classification. Over the years, several methods have been developed for the definition of local zones and location-based pruning [36]. These are often based on external data [37], incorporated into existing unmixing techniques [38], [39] or based on object-based classification [36], [40].

This article addresses the underexplored topic of how pruning affects spectral library composition. It also stresses the need for more hybrid pruning workflows that combine the strengths of different types of library pruning. We propose a new hybrid pruning approach, based on MUSIC, called multiple signal classification–confusion reduction (M-CORE). M-CORE is designed for unmixing contexts, meaning for mapping efforts in which the aim is to predict LC fractions representing subpixel composition instead of discrete per-pixel labels. The method is compared with MUSIC and AMUSES, as all three approaches rely on the same image-based component. Moreover, M-CORE is combined with ISD, to evaluate the added benefit of distance-based pruning on a GSL. The aim of this article is to: 1) assess the composition of a GSL before and after applying different pruning workflows; 2) evaluate the ability of the differently pruned GSLs to be successfully leveraged as training data for regression-based mapping of vegetation-impervious-soil (VIS) subpixel LC fractions; and 3) assess to what extent GSL-based mapping is feasible when no local spectra are available, as is often the case in real-world applications. A provisional GSL, one of the largest spectral libraries applied in urban remote sensing research to date, has been developed to perform these experiments. The proposed GSL contains spectra from the cities of Brussels, Dresden, Potsdam, Munich, Berlin, London, and Santa Barbara, as well as spectra from other published libraries mainly representing nonurban materials, and were obtained from imaging and lab spectroscopy.

II. Study Area, Data, and Preprocessing

A. Study Area

The study area covers the Brussels capital region (BCR) and parts of the neighboring region of Flanders, both located in Belgium. Fig. 1 shows brightness values for the Sentinel-2 image of the area used in this study. The city center is located in the western part of the image. In the Southeast, the large Sonian forest can be seen. Brussels International

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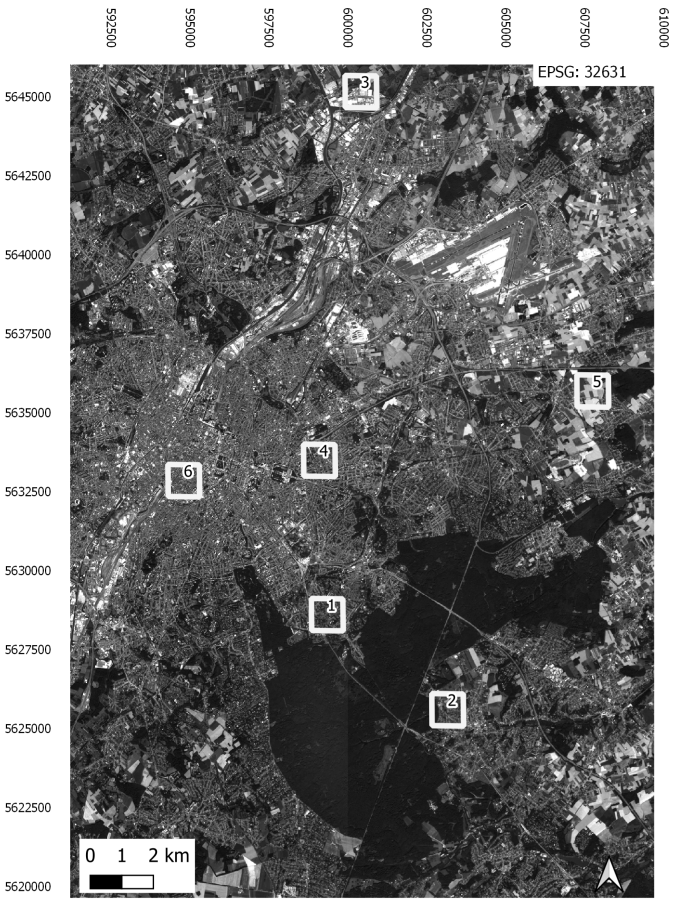


Fig. 1. Sentinel-2 image of the study area. Validation areas correspond to different types of urban morphology and contain a representative mix of LC fabrics occurring in the study area: 1) sparse urban fabric; 2) suburban; 3) industry; 4) medium-dense urban fabric; 5) rural; and 6) dense urban fabric [41].

Airport, located in the municipality of Zaventem, is visible in the northeastern part of the image, and can be recognized by its large and bright roofs and runway areas. The canal that runs diagonally from the Southwest to the Northeast of Brussels can be recognized by its low reflectance values. Most industrial activities in Brussels are located around the canal, to the North and South of Brussels. The urban fabric is characterized by a strong mix of artificial materials and urban vegetation. In the rural surroundings of the city, agricultural activity and forest are the dominant land use types.

B. Imagery and Reference Data

A Sentinel-2 reflectance image of August 22, 2015 was used as image data (Fig. 1). Reference data from [41] defining 20-m resolution VIS fractions were used as ground truth. In Fig. 1, six validation areas are shown. Each area corresponds to a different type of urban morphology representing sparse urban fabric, suburban, industry, medium-dense urban fabric, rural, and dense urban fabric. The validation areas are as such considered representative for the strong mix of urban LC in the Brussels image.

C. Provisional GSL

For the purpose of this study, a "provisional GSL" was developed that contains six spectral libraries, including one on-site library containing spectra from Brussels and five libraries of spectra that were sampled off-site. The on-site Brussels library contains 920 spectra, sampled from a 2-m resolution Airborne Prism EXperiment (APEX, [42]) image of Brussel. Off-site libraries included in the GSL are: a HUB library containing 221 hyperspectral mapper (HyMap, [43]) spectra covering Berlin [8], a German Aerospace Center—Deutsches Zentrum für Luft- und Raumfahrt (DLR)—library containing 4531 HyMap spectra representing basic urban LC types from Dresden, Potsdam, and Munich [9], and an ECOSTRESS library containing 1045 spectra from mostly natural material types around the globe. Spectra were acquired from airborne image data, by the ECOSTRESS radiometer and by additional laboratory measurements [1], [4]. Also, a SLUM library, containing 68 impervious surface spectra derived using measurements made by a portable Fourier transform infrared (FTIR) spectrometer containing London urban construction materials, such as lead and iron roofs, was included [10]. Finally, a University of California, Santa Barbara (UCSB) library, containing 850 spectra from the area near Santa Barbara, California (U.S.), was added to the GSL. The UCSB spectra were acquired *in situ* during the spring of 2001, using an analytical spectral device's (ASD) full-range spectrometer [3]. An overview of the key features of the original libraries is given in Table I.

D. Preprocessing

The Sentinel-2 image encompassing the study area was atmospherically corrected using Sen2Cor to acquire surface reflectance values [44]. Nine Sentinel-2 bands (2–7, 8a, 11, and 12) covering spectral intervals with high atmospheric transmission were retained. The image data were bilinearly interpolated to 20-m resolution and coregistered to the reference data pixels using OSGeo4W Shell resampling commands.

Reference LC fractions from [41] were produced with two approaches, depending on the location of the validation area. For validation areas 1, 4, and 6, which are located within the BCR, an overlapping 2-m resolution APEX hyperspectral image, acquired in the summer of 2015, was classified using support vector classification (SVC). Geometric features, such as terrain slope and height, derived from a high-density light detection and ranging (LiDAR) point cloud were incorporated to improve mapping accuracy through postclassification correction [45]. Reference data for validation areas 2, 3, and 5, located in the Flemish part of the study area, were obtained from a 2012 high-resolution LC map, produced by Informatie Vlaanderen (Flanders Information Agency). It was confirmed that little to no LC change occurred between 2012 and 2015 in the chosen validation areas.

Because the spectral libraries included in the proposed GSL were produced with different sensors, they are characterized by different band set definitions. All library spectra were thus resampled to the nine selected Sentinel-2 bands using the Gaussian spectral convolution. From these spectral libraries,

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TABLE I
DESCRIPTION OF ORIGINAL GSL SPECTRAL LIBRARIES

Library	Sensor/ spectrometer	Site	Spectra	Bands	Bandwidth (nm)	Spatial resolution (m)	Source
Brussels	APEX	Brussels	920	206	3.4 – 11.4	2.0	F. Priem (unpublished)
HUB	HyMap	Berlin	221	111	14.5	3.6	[8]
DLR	HyMap	Dresden, Potsdam and Munich	4531	113	15.0	3.3 – 7.0	[9]
ECOSTRESS	Spectrometer	Global	1045	258	7.7	/	[4]
SLUM	Spectrometer	London	68	988	2.2	/	[10]
UCSB	Spectrometer	Santa Barbara	850	849	2.0	/	[3]

two GSLs were created. The first library, from now on referred to as GSL_B (GSL with local Brussels spectra), includes all libraries. GSL_{NB} (GSL without local Brussels spectra) on the other hand includes only off-site spectra, i.e., all but the Brussels library. The generic class hierarchy of the GSL is shown in Table II. Vegetation spectra are subdivided into nonphotosynthetic vegetation, woody vegetation, herbaceous vegetation, and green algae. Fine-grained mineral deposits include soil and sand spectra. Impervious spectra are subdivided into mineral pavements, mineral roofs, ceramic pavements, ceramic roofs, polymer pavements, polymer roofs, metal roofs, and specific structures.

III. METHODOLOGY

A. M-CORE

A new local library pruning method called M-CORE is proposed. It builds on the MUSIC algorithm [31] and is specifically developed for LC fraction mapping applications. The method consists of different steps that are illustrated in Fig. 2. The first step involves segmenting the image in areas of similar LC composition based on identifying one or more dominant class(es). A description of the approach used for producing this initial segmentation in our case study is included in Section III-E. Then, for each image segment, the image-based MUSIC approach is applied on the library, in which spectra below a predefined MUSIC distance threshold are retained. The MUSIC distance is calculated by taking the Euclidean distance between a library spectrum and the subspace formed by the first couple of eigenvectors of the covariance matrix of the image segment [31]. Spectra that may cause spectral confusion between different LC classes in the segment due to spectral similarity are then targeted by the CORE component that draws on aspects of distance- and image-based pruning. CORE starts by constructing a pairwise distance matrix between all remaining spectra in the MUSIC-pruned GSL obtained for each of the segments. L1-sam, which combines spectral angle mapper (SAM) [46] with L1 distance, is used as a hybrid metric for spectral dissimilarity as in [9] and [47]

$$l1-\text{sam} = \sum_{i=1}^{n} |a_i - b_i| \sin\left(\arccos\left(\frac{\sum_{i=1}^{n} a_i b_i}{\sqrt{\sum_{i=1}^{n} a_i^2}\sqrt{\sum_{i=1}^{n} b_i^2}}\right)\right) \quad (1)$$

where a and b are the reflectance values for a spectral pair and n is the number of bands. Potential cases of spectral confusion, i.e., pairs of candidate spectra belonging to different LC classes with a spectral distance lower than a predefined threshold, are iteratively evaluated. Pairs with the lowest pairwise distance are evaluated first. For each such pair of spectra, a spectrum belonging to one of the dominant LC classes in the corresponding segment is retained, while a spectrum not belonging to one of the dominant classes is discarded. This way, the method aims to discard nondominant spectra that may cause spectral confusion with dominant spectra. If both evaluated spectra belong to a dominant class of the segment, both are retained. If neither of the evaluated spectra belong to a dominant class of the segment, both are retained as well.

B. Library Pruning Experiments

Different pruning setups were defined based on state-of-the-art library pruning approaches. These setups were compared to assess the added value of the newly proposed M-CORE approach in predicting VIS LC fractions using multisite spectral libraries. The tested pruning methods are MUSIC, AMUSES, and the proposed M-CORE approach. All three methods have an image-based component. AMUSES and M-CORE combine image- and distance-based pruning. In an additional experiment, M-CORE is combined with ISD, to assess the added value of combining the method with a distance-based pruning approach that only considers spectral similarity between pairs of spectra, irrespective of image context.

An overview of the different pruning setups compared in this study is given in Table III. All experiments start with a GSL, except experiment 1 which uses the Brussels library only. This baseline scenario enables us to compare the traditional use of a site-specific library with the proposed use of a GSL including spectra collected elsewhere. All experiments apply library pruning prior to mapping, except experiments 1–3. GSL experiments start from an initial GSL including or not including spectra collected within the Brussels study area (abbreviated by subscript B and NB, respectively). The assumption is that the presence of local spectra in the GSL will improve mapping accuracy, and that image-based pruning methods will select these local spectra. On the other hand, spectra from other sites may complement a local library, particularly if some locally occurring cover types are not well represented in this library. Including certain nonlocal spectra

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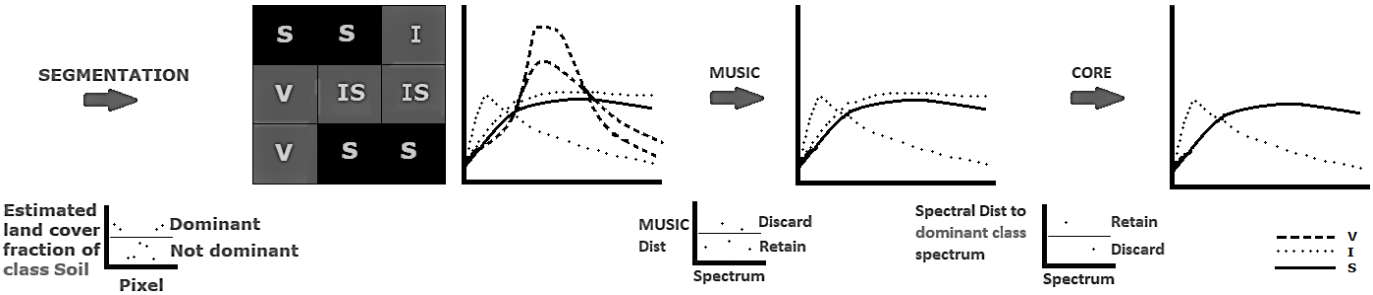


Fig. 2. Concept of M-CORE. M-CORE starts from an initial LC fraction map (in our case study, a VIS map). Pixels in this preliminary map are allocated to image segments with the same dominant LC composition based on minimum LC fraction thresholds, defining if a class is dominant or not. Then, for each image segment, the image-based MUSIC approach is applied to retain EMs from the spectral library that are similar to spectra found within the image segment considered. The CORE component then iteratively evaluates spectral distance between the closest spectra belonging to different classes. If the pairwise spectral distance is below a predefined distance threshold, and one of the spectra belongs to a dominant class, while the other spectrum does not, then the nondominant spectrum is discarded. This way, only nondominant spectra that are significantly different from dominant spectra are retained.

may therefore improve the quality of the mapping. MUSIC and AMUSES were applied using a global (abbreviated by subscript GLO) as well as a local pruning approach (abbreviated by subscript LO). Global pruning implies that the algorithm is applied once, on the entire image, while local pruning means that the algorithm is applied separately on each image subset, consisting of pixels with similar class composition (see Section III-E for more information on image segmentation). We expect that local methods would be able to select spectra from the GSL that better represent the materials that are present at the local level and thus improve the mapping. Finally, experiments 12–15 encompass the experiments that include M-CORE and that are by default applied locally. We expect that the experiments that include M-CORE would yield better mapping results than prior experiments because of the method's aim to address interclass spectral ambiguity in the mapping process, a problem that we expect to be a major source of GSL-based mapping errors. In experiments 14 and 15, M-CORE is combined with ISD, to assess the added value of combining the method with distance-based pruning. This is expected to decrease the amount of retained spectra even more. The pruned GSLs were compared based on their composition and their ability to be leveraged as training data for LC fraction mapping. This is discussed in Sections III-C and III-D.

C. Effect of Pruning on GSL Composition

GSL composition before and after running the different pruning experiments was assessed in terms of fraction of retained spectra originating from the different spectral libraries included in the GSL and in terms of the abundance of the spectra representing each material type. For this analysis, we limit ourselves to experiments 1, 2, 4, 6, 8, 10, and 12.

D. Effect of Pruning on VIS Fraction Estimation

To assess the ability of the pruned libraries to be leveraged as training data for LC mapping, the libraries were used as input for estimating subpixel VIS fractions from a Sentinel-2 image covering Brussels. The workflow for VIS fraction estimation is described in Fig. 3. For each experiment, the resulting library spectra were first synthetically mixed to obtain mixed spectra for different class fraction combinations, which could then be used as training data for regression-based class fraction estimation. Synthetic mixing was applied here by randomly sampling EM spectra from the library and combining them bandwise, in a linear fashion, to produce spectral signatures representing different mixtures of LCs [2], [48]. We generated both binary (two source spectra) and tertiary (three source spectra) mixtures. To address the strongly differing number of spectra per class in the GSL, stratified sampling was used to randomly generate the synthetic mixtures. This ensures that the mixed training data equally represents all possible class mixtures [49]. For each global pruning experiment, and for each set of local spectra in the local pruning experiments, 1000 synthetic mixtures were randomly generated using the corresponding spectral library. Then, the synthetic mixtures were used as training data to fit support vector regression (SVR) models for estimating the fraction of the three VIS classes. For this research, we used ϵ-SVR as implemented in the Python-based Scikit-Learn support vector machine (SVM) package [50]. The regularization, kernel, and epsilon loss parameters were optimized with a grid search [51]. The trained SVR models were applied on the Sentinel-2 image of Brussels to predict VIS fractions for each pixel. Predicted fractions smaller than 0 or larger than 1 were, respectively, replaced with values 0 and 1. Because we use separate SVR models for each class, the sum of class fractions per pixel may not sum to 1. However, for the majority of the pixels, only small deviations from 1 were observed

For each experiment, predicted VIS fractions were validated against reference fractions. Validation was performed on a 3×3 pixel basis, i.e., by spatially resampling fraction maps to the level of cells covering 3×3 base resolution pixels. This reduces the impact of small coregistration errors between reference data and fraction estimates on the accuracy assessment. Mean error (ME) and mean absolute error (MAE) were used as validation metrics

$$\mathrm{ME} = \frac{1}{n}\sum_{i=1}^{n}(f_{\mathrm{pred}_i} - f_{\mathrm{ref}_i})$$
$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|f_{\mathrm{pred}_i} - f_{\mathrm{ref}_i}| \quad (2)$$

where n is the total amount of 3×3 pixel validation blocks, f_{pred} is the predicted LC fraction, and f_{ref} is the reference LC fraction.

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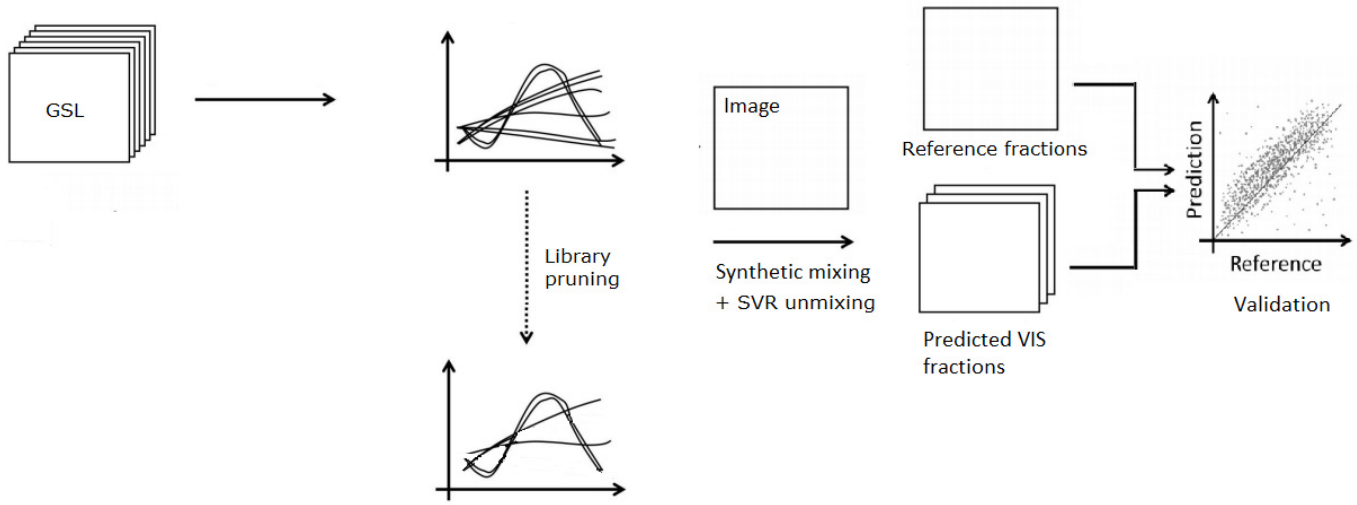


Fig. 3. Main methodological steps for the mapping experiments. GSL spectra (after pruning) are synthetically mixed. The obtained mixtures are used as training data for SVR unmixing. Unmixing results in predicted VIS fractions, which are then validated against reference fractions.

E. Initial Segmentation

As explained in Section III-B, all tested pruning approaches were applied both globally and locally, with exception of M-CORE, which is inherently local. In the global pruning approach, the GSL is optimized for use on the full image. In the local approach, several instances of pruned GSLs are generated that are specifically optimized for use on different parts of the image. Because, in the experimental setup of this study, we chose to focus on VIS mapping from Sentinel-2 data, image partitions were defined as in [39], i.e., by segmenting the image based on a preliminary estimation of per-pixel VIS fractions, where pixels were grouped based on class composition. Just as for the VIS mapping based on the pruned libraries (Section III-D), subpixel VIS fractions for this preliminary mapping were estimated using SVR, yet this time with training data generated by synthetic mixing of spectra from a GSL obtained after global pruning of the original multisite library with MUSIC. The $\text{MUSIC}_{\text{GLO}}$-percentage-retained threshold used for the image segmentation was set to 50 (for more details on the SVR training process, see Section III-D). Based on VIS class fractions, image segments for local library pruning were defined by identifying the dominant classes within each pixel. Pixels were considered as dominated by one class if that class covered at least 2/3 of a pixel. Pixels were considered as dominated by two classes if the combined fraction coverage of these classes was at least 80%, with a minimum of 40% each. This segmentation yielded seven segments, each corresponding to one, two, or no dominant classes, i.e., V, I, S, vegetation-impervious (VI), vegetation-soil (VS), impervious-soil (IS), and VIS.

F. Parameterization of Pruning Approaches

Each of the pruning experiments requires parameterization. For each experiment, out of a discrete set of possible parameter values, the set was chosen that yielded the best mapping results in terms of MAE of predicted VIS fractions against reference fractions. Experiments were performed multiple times, in order to increase the reliability of parameterization. Optimal parameter values for the different experiments are described in Table IV (Appendix A).

For MUSIC, a fraction of the spectra with the lowest MUSIC distance is selected. This is indicated by a percentage-retained threshold. For $\text{MUSIC}_{B,\text{LO}}$ and $\text{MUSIC}_{\text{NB,LO}}$, only 42.5% of library spectra had to be retained for optimal mapping results. This is lower than the global threshold of 50% used for $\text{MUSIC}_{B,\text{GLO}}$ and $\text{MUSIC}_{\text{NB,GLO}}$, and may be related to spectrally more uniform image segments that can be described with fewer spectra.

For AMUSES, a fraction of the spectra with the lowest MUSIC distance is selected (%retain) and a fraction of the spectra with the highest MUSIC distance is discarded (%remove). As in [12], the remaining spectra that are neither selected nor discarded are then evaluated based on their distance to the retained spectra, using the Jeffries–Matusita distance-SAM (JMD-SAM) metric, and are included in the library only if they are sufficiently different from already retained spectra. The JMD-SAM distance threshold used is spectrum-specific though and increases the more the spectrum differs from the initially retained spectra. As such, spectra differing too strongly from retained spectra are less likely to be selected. The spectrum-specific threshold lies between a minimum and maximum threshold value $\text{thold}_{\text{low}}$ and $\text{thold}_{\text{high}}$ that needs to be defined by the user and linearly increases between both values depending on the MUSIC distance of the spectrum. Optimal results for experiments with AMUSES were obtained with the following pruning parameters: %retain $=$ 0.05, %remove $=$ 0.2, $\text{thold}_{\text{low}} = 0.00001$, and $\text{thold}_{\text{high}} = 0.001$.

As discussed in Section III-A, M-CORE requires the choice of a MUSIC threshold for MUSIC_{LO} pruning of the spectral libraries of the image segments, as well as one L1-sam distance threshold for the CORE component. Optimal results for M-CORE result in a percentage-retained threshold of 50 and an L1-sam distance threshold of 0.01. For the M-CORE experiments including ISD, L1-sam distance threshold of 0.0001 was used.

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TABLE II
GENERIC CLASS HIERARCHY OF GSL_B

VIS	Material type level 1	Material type level 2
V	Non-photosynthetic vegetation	Non-photosynthetic vegetation
		Plant litter
		Thatch roof
		Wood
	Woody vegetation	Broadleaf vegetation
		Coniferous vegetation
	Green algae	Algea bloom
	Herbaceous vegetation	Crop
		Extensive green roof
		Flowers
		Lawn
		Lichen
		Meadow
		Pre-senescent crop
		Semi-intensive green roof
		Senescent crop
		Succulent vegetation
I	Mineral pavements	Cobble pavement
		Gravel pavement
	Mineral roofs	Gravel roof
		Slate roof
	Ceramic pavements	Clay ceramic pavement
		Concrete pavement
	Ceramic roofs	Asbestos fiber cement roof
		Cellulose fiber cement roof
		Clay ceramic roof
		Concrete roof
		Glass roof
	Polymer pavements	PE pavement
		PU pavement
		Asphalt pavement
		Artificial turf
	Polymer roofs	EPDM roof
		PC roof
		PE roof
		PP roof
		PVC roof
		Acrylic glass roof
		Bitumen roof
		Modified bitumen roof
		Tar paper roof
	Metal roofs	Aluminium roof
		Copper roof
		Galvanume roof
		Galvanized steel roof
		Iron roof
		Lead roof
		Metal roof
		Steel roof
		Zinc roof
	Specific structures	Railroad track
		Rock
		Roof-mounted solar panel
S	Fine-grained mineral deposits	Sand
		Soil

TABLE III
OVERVIEW OF PRUNING SETUPS APPLIED IN THE MAPPING EXPERIMENTS

Nr.	Abbreviation	Pruning?	Brussels spectra?	Scale of pruning
1	Brussels	No	n/a	n/a
2	GSL_B	No	Yes	n/a
3	GSL_{NB}	No	No	n/a
4	$MUSIC_{B,GLO}$	Yes	Yes	Global
5	$MUSIC_{NB,GLO}$	Yes	No	Global
6	$MUSIC_{B,LO}$	Yes	Yes	Local
7	$MUSIC_{NB,LO}$	Yes	No	Local
8	$AMUSES_{B,GLO}$	Yes	Yes	Global
9	$AMUSES_{NB,GLO}$	Yes	No	Global
10	$AMUSES_{B,LO}$	Yes	Yes	Local
11	$AMUSES_{NB,LO}$	Yes	No	Local
12	$M\text{-}CORE_B$	Yes	Yes	Local
13	$M\text{-}CORE_{NB}$	Yes	No	Local
14	M-CORE + ISD_B	Yes	Yes	Local
15	M-CORE + ISD_{NB}	Yes	No	Local

IV. RESULTS AND DISCUSSION

A. Effect of Global Pruning on Library Composition

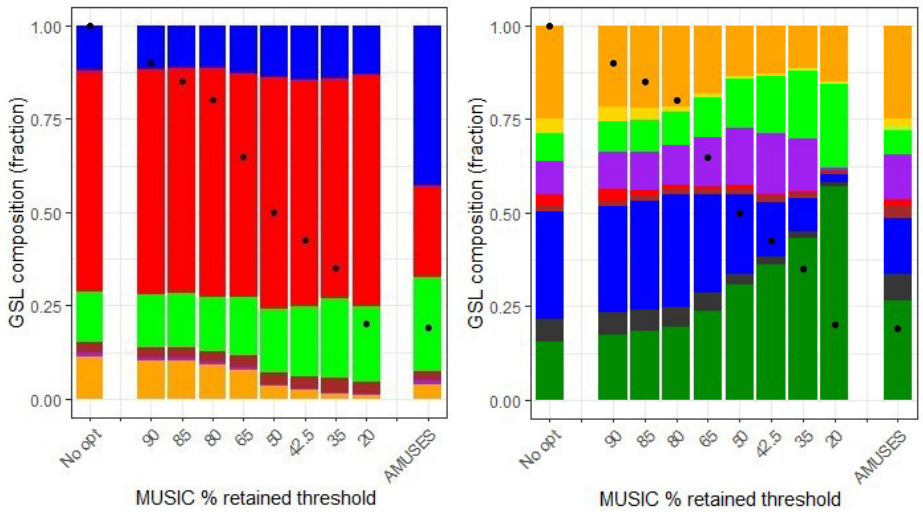
Fig. 4(a) and (b) shows the composition of the GSL including Brussels spectra (GSL_B) before and after each global pruning experiment. The resulting library composition is described in terms of the relative frequency of spectra retained from the original spectral libraries [Fig. 4(a)] and the relative frequency of material type spectra [Fig. 4(b)]. For $MUSIC_{B,GLO}$, the impact of different values for the threshold percentage of spectra retained is also shown. The relative size of the pruned libraries with respect to the initial size of the GSL is indicated with black dots.

As can be seen in Fig. 4(a), in the initial GSL before pruning, most spectra come from the DLR library. The initial GSL consists mainly of ceramic and polymer pavement and roof spectra, whereas a smaller fraction describes other material types, such as mineral pavements and roofs, green algae, and nonphotosynthetic vegetation [Fig. 4(b)]. The different globally applied pruning approaches affect the GSL in different ways. Image-based approaches like MUSIC and AMUSES are expected to retain a set of spectra that optimally describes the spectral variability that is present in the target image. Applying $MUSIC_{B,GLO}$ on the GSL leads to a decrease in the relative presence of the UCSB library, and to an increase in the share of spectra from the ECOSTRESS, and to a lesser extent in the DLR and the Brussels libraries. As shown in Fig. 4(b), MUSIC pruning also leads to considerable changes in the relative presence of material types. Material types that are common in the Brussels image, such as ceramic pavements and roofs, woody and herbaceous vegetation, are clearly retained more often than other material types. The large area covered by the Sonian forest and the agricultural plots around the city may explain the larger share of vegetation spectra. $AMUSES_{B,GLO}$ combines image- and distance-based pruning. In our case, applying $AMUSES_{B,GLO}$ leads to a relatively small spectral library of 1460 spectra, clearly more dominated by the Brussels site-specific spectra than that is the case for the original GSL. In terms of material composition, the relative frequencies of spectra for different groups of materials remain closer to the original GSL_B than that is the case with $MUSIC_{B,GLO}$, for a similar library size, resulting in a higher within-class spectral diversity of the pruned GSL.

B. Effect of Local Pruning on Library Composition

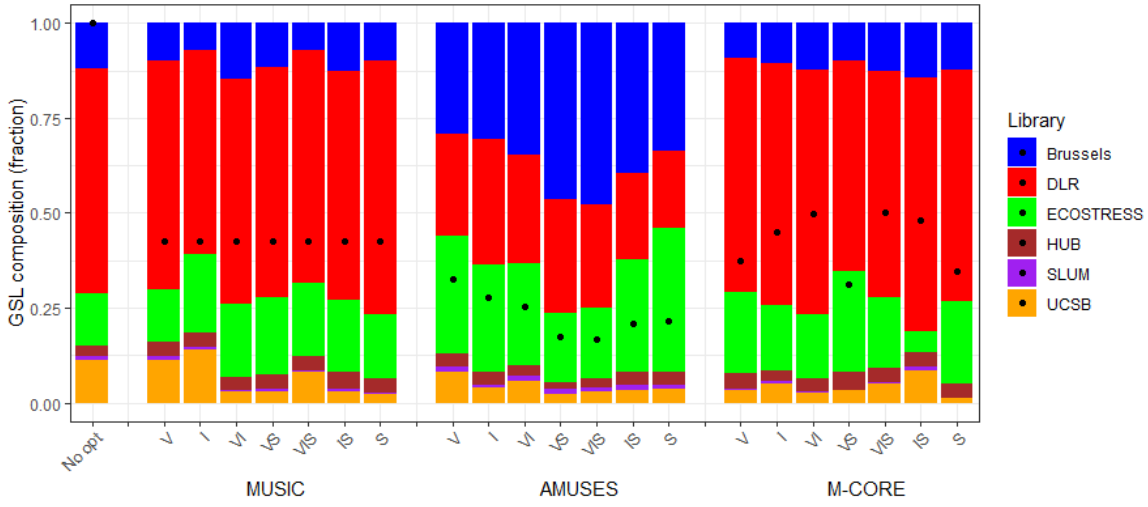
As explained in Section III-E, local pruning requires the delineation of local zones in the target image, which, in our case, were defined based on a preliminary assessment of per-pixel LC class composition (not shown). In this initial segmentation, most city center pixels are dominated by

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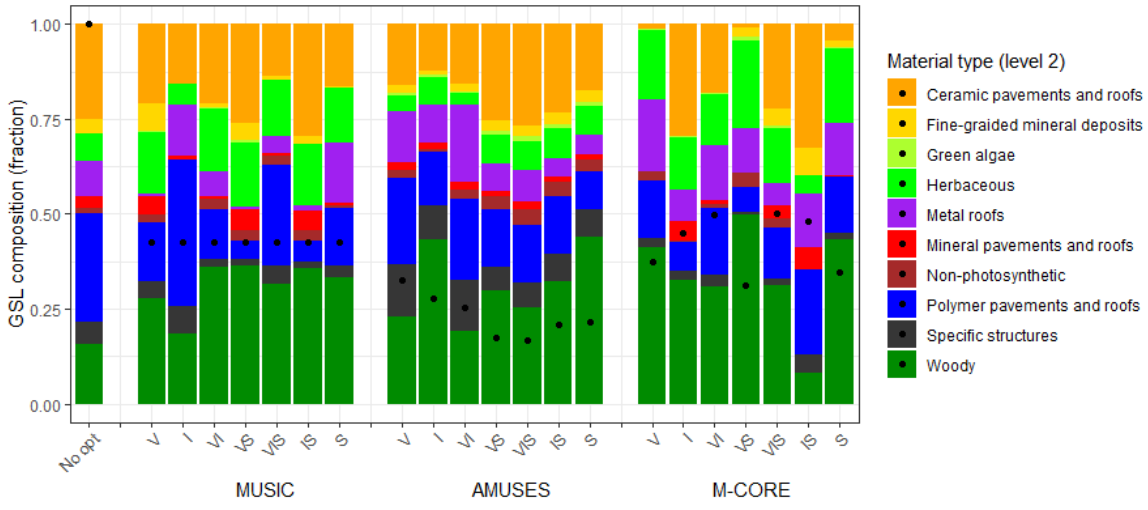


(a)

(b)



(c)



(d)

Fig. 4. Relative frequency distribution of spectra of different (a) and (c) spectral libraries and (b) and (d) material types in the GSL before and after (a) and (b) global or (c) and (d) local pruning. Black dots indicate the relative size of the pruned libraries with respect to the GSL's original size.

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impervious surface cover. In suburban areas, dominant LC is mostly mixed (VI, IS, and VIS).

Fig. 4(c) and (d) shows the composition of the GSL including local spectra for the three image-based pruning techniques, i.e., $\text{MUSIC}_{B,\text{LO}}$, $\text{AMUSES}_{B,\text{LO}}$, and the proposed M-CORE_B approach, after local pruning. Again, composition is described in terms of the relative frequency of spectra retained from the original spectral libraries [Fig. 4(c)], as well as in terms of material type composition [Fig. 4(d)].

Just as in the global pruning experiments, the library composition for $\text{MUSIC}_{B,\text{LO}}$ shows a stronger resemblance with the original GSL_B composition than is the case for $\text{AMUSES}_{B,\text{LO}}$, where the site-specific spectra for Brussels clearly have a larger share than in the original generic library, while the share of UCSB and DLR fractions is lower [Fig. 4(c)]. Overall, the M-CORE_B GSLs have a similar library composition as those obtained by applying $\text{MUSIC}_{B,\text{LO}}$. For all the three methods, an increase in the relative share of ECOSTRESS spectra is observed. If we compare the library composition for specific segments of the image for each of the three pruning methods, we notice small shifts in the relative share of the different libraries depending on LC composition, yet no clearly interpretable patterns are observed. If we analyze the relative frequencies of material types in the pruned GSLs, we notice that, in all three approaches, the relative share of woody spectra is higher than in the original library. This may also explain the more dominant presence of ECOSTRESS spectra in the pruned GSLs compared with the original GSL.

When applying $\text{AMUSES}_{B,\text{LO}}$, we observe that some segments (e.g., VIS and VS) are more strongly pruned than others (e.g., V and I). This implies that, for the VIS and VS segments, less spectra are retained between the 5% retain and 20% remove thresholds than for the other segments, indicating that less of these spectra are sufficiently dissimilar from the initially retained 5% lowest MUSIC distance spectra. We notice the opposite in image segment V, resulting in a relatively large amount of nonvegetation spectra in the pruned GSL. Similarly, in image segment I, a relatively large amount of nonimpervious spectra is retained.

Contrary to $\text{AMUSES}_{B,\text{LO}}$, M-CORE_B's distance-based component causes less pruning of GSLs for image segments with two dominant LCs (VI and IS), because more dominant classes implies less pruning of nondominant class spectra by the CORE component. If no dominant class is present (VIS), the CORE component will not remove any of the spectra obtained after applying the MUSIC algorithm. For VS, however, even with two classes being dominant, many spectra seem to be discarded due to significant spectral similarity between nondominant impervious spectra and dominant soil spectra.

C. Effect of Pruning on VIS Fraction Estimation

Sections IV-A and IV-B discussed the effect of pruning on GSL composition. To assess how pruning affects a library's ability to accurately map LC, we used the pruned libraries as input for generating training data, for VIS LC fraction estimation with SVR. Validation results for the experiments with and without library pruning are summarized in Fig. 5 and Table V (Appendix B). The estimated VIS fractions for the six validation blocks for experiments 1, 2, 4, and 12, visualized as false color composites, are shown in Fig. 6. Fig. 7 displays scatterplots of reference versus estimated VIS LC fractions for use of the multisite library without pruning (GSL_B) and for use of the library when applying M-CORE (M-CORE_B).

Before considering the effects of pruning, we see that, for impervious surfaces and soils, fraction mapping accuracies obtained with a GSL are lower than when using a single-site library that only contains spectra from Brussels. Similar results are reported in other studies using multisite libraries [22], [41]. Additionally, we observe that GSL_B containing Brussels spectra yields better results than the GSL not including these local spectra. This also concurs with [22] and can be explained by the broad and often site-specific variety of materials, especially materials used for road, pavement, and roof construction. The spectral variability of these material types may be insufficiently represented by spectra that are collected off-site. On the other hand, as more spectra are added to a multisite GSL, confusion between spectrally similar soil and impervious surface materials becomes more likely and this also affects mapping accuracy [41], [52], [53]. As shown in Fig. 5, fraction estimates are worse for impervious surface types and soil than for vegetation. There is also a substantial positive and negative bias in fraction estimates for impervious surfaces and soil, respectively. This can again be explained by the strong spectral confusion between these classes.

Pruning the GSL generally yields improved fraction mapping accuracies for all three VIS classes (Fig. 5). More importantly, pruned GSLs tend to yield similar or, in some cases, even better fraction estimates than when using the Brussels library on its own. Furthermore, it is interesting to see that accuracy differences between the use of GSLs with and without local spectra (*B*, NB) decrease when applying pruning.

Applying MUSIC on the GSLs leads to considerable improvements in MAE for impervious surfaces and soil compared with the no-pruning GSL results. Applying MUSIC locally yields similar results as applying it globally. Closer inspection of the validation areas reveals that pixels assigned to the correct image segment in the segmentation process yield better fraction estimates compared with global pruning. On the other hand, pixels that are not assigned to the correct image segment tend to yield worse fraction estimates, mostly due to spectral similarity between specific impervious surface and soil materials. Applying AMUSES leads to similar improvements as with MUSIC and thus, in our study, fails to improve mapping accuracies relative to MUSIC, mostly due to unwarranted pruning of spectra. Also for AMUSES, accuracies obtained with local and global pruning are similar.

Both MUSIC and AMUSES suffer from spectral confusion, especially between soil and impervious, leading to a general overestimation of soil and underestimation of impervious surface cover (Fig. 5). The M-CORE approach introduced in this article addresses the problem of working with large multisite libraries including spectrally similar EMs belonging to different material classes. It first aims to retain image-relevant

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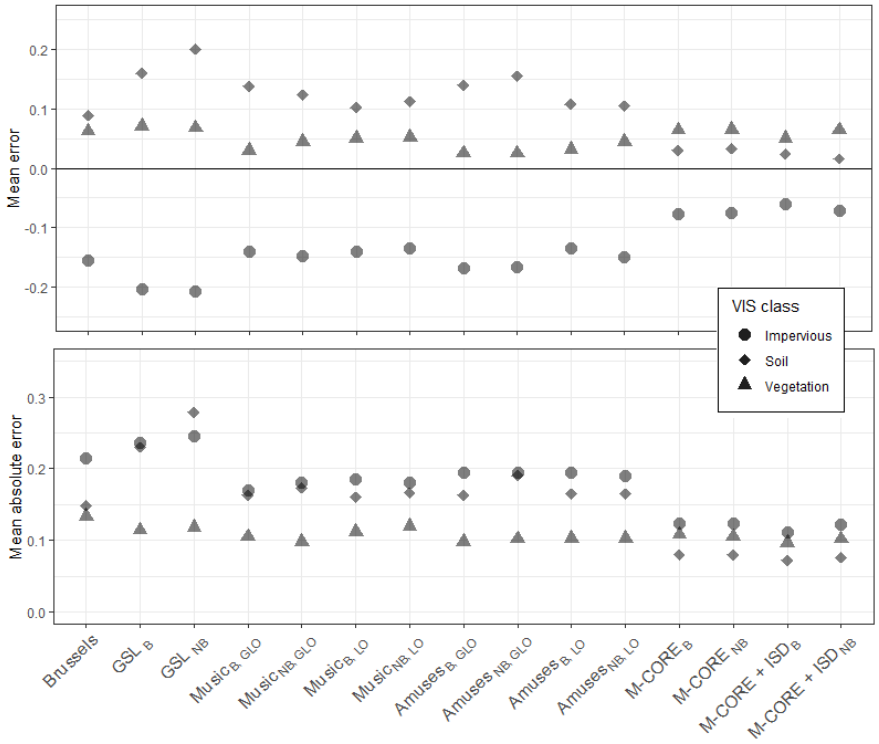


Fig. 5. ME and MAE for all mapping experiments for each VIS class.

spectra by applying MUSIC_{LO}. Then, it deals with spectral confusion through its CORE component. In terms of LC fraction accuracies, M-CORE yields the best results of all tested methods (Fig. 5). Most noticeable is the reduction in bias and absolute error for both low and high impervious and soil fraction estimates (Fig. 7). Nevertheless, also when applying M-CORE, we still observe that, for part of the pixels, rest fractions are estimated of classes that are not actually present within the pixel, particularly for soil. These rest fractions are mainly caused by the presence of materials in the pixel that are spectrally similar to materials belonging to another class. Spectral confusion may also lead to the assignment of a pixel to the wrong segment in the initial segmentation phase. In that case, M-CORE may retain a nonrelevant spectrum and discard a relevant one.

Extending M-CORE with distance-based pruning (ISD) results in mapping accuracies that are similar to applying M-CORE without ISD. However, the amount of retained spectra decreases from 31% to 50% of the initial GSL size for M-CORE without ISD to 13%–24% for M-CORE with ISD. This results in lower computational demand during the unmixing process without loss of performance and is an interesting feature for GSL-based mapping workflows.

While our results show that M-CORE manages to significantly limit spectral confusion and hence improve LC fraction mapping, like all pruning methods, it is important that parameter values are properly chosen in order to obtain optimal mapping result. The first set of parameters that need to be set are the parameter values required for segmentation. In this article, we chose for user-defined LC fraction thresholds to assign pixels to a particular image segment. Defining too low fraction thresholds for considering a class as dominant within a pixel increases the chance of mistakenly assigning a pixel to a particular segment (false positive), while the reverse occurs when defining too high fraction thresholds (false negative). Our experiments showed that allocating pixels to the correct segment is of high importance for obtaining accurate mapping results. The other two parameter values are the MUSIC_{LO} and CORE pruning parameters. Choosing an optimal MUSIC_{LO} threshold is important to retrieve relevant library spectra. With a too low threshold spectra that would be relevant to include might be discarded. A too high threshold may result in a pruned library that is not sufficiently optimized for the image segment it is used for. Choosing an optimal CORE distance threshold is important for discarding spectra that may potentially cause spectral confusion. A too low distance threshold results in a too small reduction in spectral confusion between retained library spectra. A too high distance threshold may result in relevant spectra being mistakenly discarded for the purpose of limiting spectral confusion.

D. M-CORE as Part of a Multistep Approach to GSL Pruning

When applying pruning on a GSL to extract an optimal set of EM spectra as training data for subsequent mapping, retained spectra should ideally be: 1) relevant for the target image; 2) nonredundant with respect to other library spectra; and 3) selected with the intent to reduce between-class spectral confusion during the mapping process. In this regard, an analogy can be made with the Swiss Cheese Model, developed by Reason [54], as seen in Fig. 8. The model was

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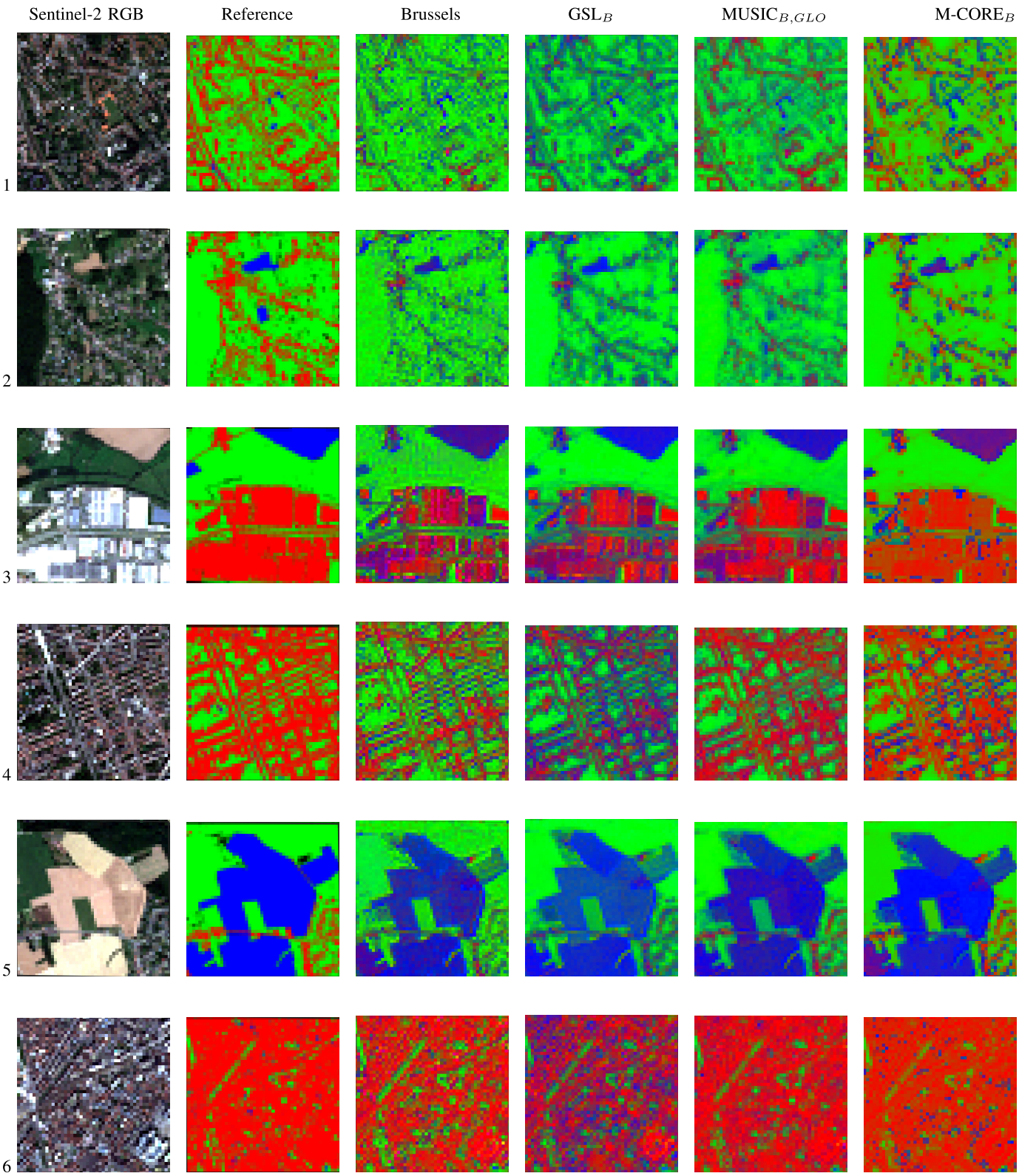


Fig. 6. Reference fractions and VIS fraction composites obtained with the different mapping experiments for the six validation sites shown in Fig. 1. Color scheme: Vegetation = green; impervious = red; soil = blue.

originally developed to conceptualize how defenses, barriers, and safeguards may be penetrated by an accident trajectory. In a similar way, when optimizing a spectral library, different approaches should ideally be combined to retain warranted spectra. In this study, different pruning approaches have been evaluated. MUSIC aims to discard spectra that are not relevant as training data. Methods such as EAR, ISD, and IES focus on discarding redundant spectra. Hybrid approaches, such as AMUSES, discard both irrelevant and redundant spectra. Often, although not by design, existing approaches

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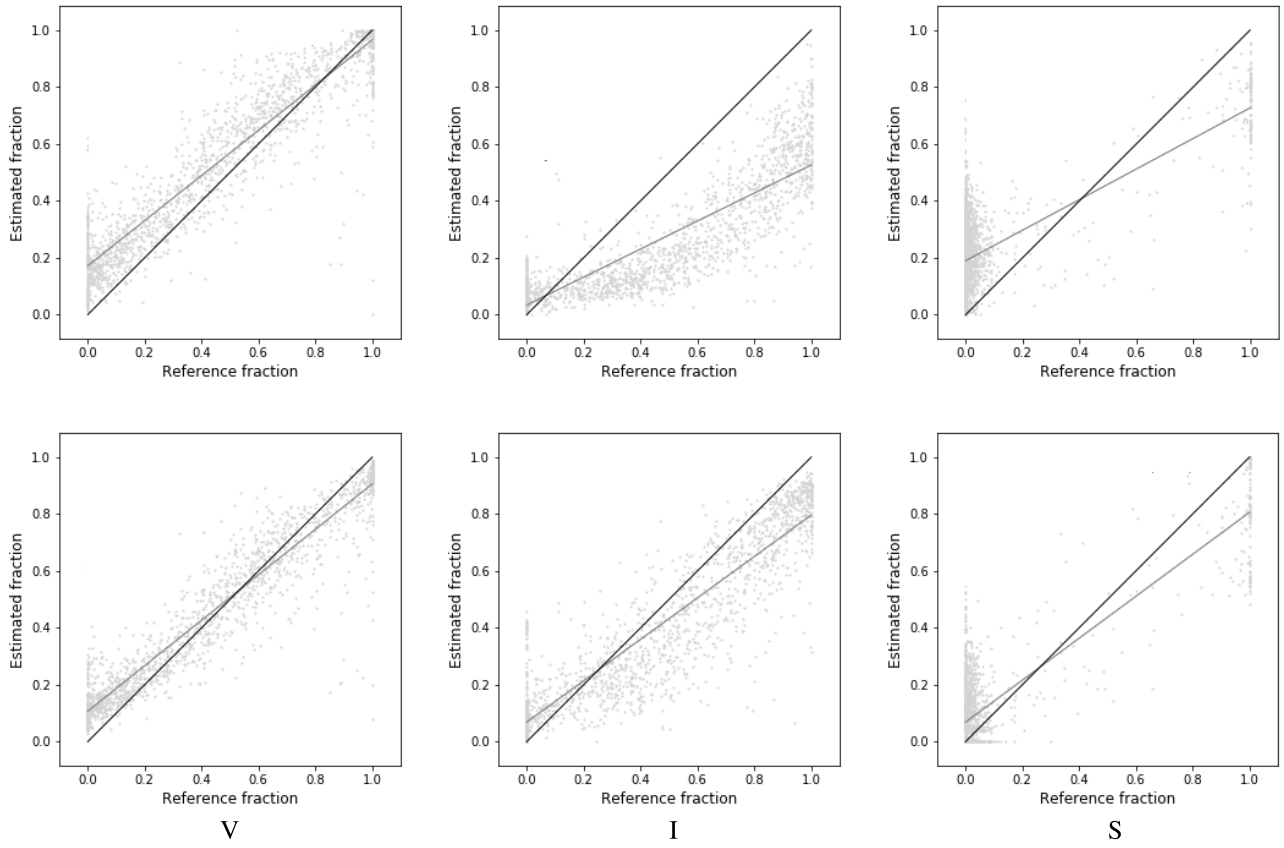


Fig. 7. Scatterplots displaying reference versus estimated VIS LC fractions for experiments 2 [GSL$_B$ (top)] and 12 [M-CORE$_B$ (bottom)].

also help to limit spectral confusion during the mapping process. However, so far, the potential of combining different pruning approaches has not been thoroughly investigated and existing techniques often focus on one or sometimes two of the above-mentioned objectives. M-CORE has been explicitly developed not only to retain relevant spectra but also to reduce spectral confusion during mapping. Combining M-CORE with ISD incorporates all layers in the Swiss Cheese Model for library pruning and has proven to be the best pruning setup in this proof-of-concept study.

E. Conclusions and Future Work

Generic spectral libraries, or collections of multisite, multisensor spectral libraries, will often contain pairs of spectra that are highly similar while representing different cover types. This spectral confusion is a limiting factor in the use of such libraries as reference data for mapping. In this article, we have proposed a new method for pruning spectral libraries that aims to improve the accuracy of LC fraction mapping, based on an extension of MUSIC. M-CORE starts from an initial segmentation of an image into segments with similar LC composition and involves two steps: 1) image-based pruning for the selection of relevant library spectra (MUSIC) for each image segment and 2) local pruning of the retained spectra to limit spectral confusion (CORE). The proposed method was compared with two existing, image-based pruning approaches, MUSIC and AMUSES, in dedicated mapping experiments

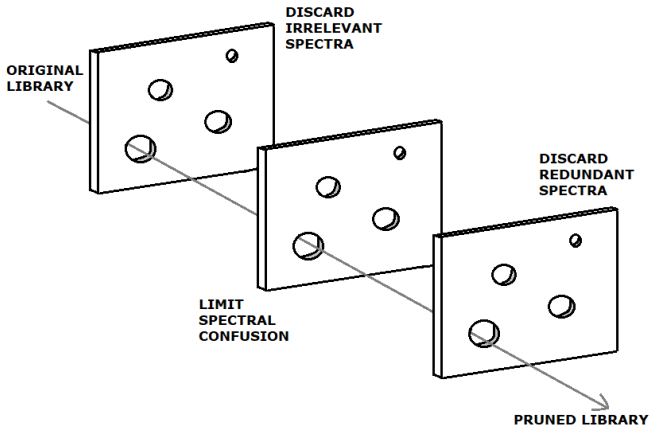


Fig. 8. Swiss cheese model for library pruning.

using a large multisite spectral library as prototype GSL. The library was created by aggregating 7635 spectra from six existing spectral libraries and is one of the largest collections of spectra used in urban remote sensing research to date. The (pruned) library spectra were leveraged as training data for SVR-based LC fraction mapping from a multispectral Sentinel-2 image of Brussels. The proposed approach for GSL pruning was shown to significantly improve mapping accuracies compared with the use of GSLs pruned with MUSIC and AMUSES. We also demonstrated that, by combining

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TABLE IV

PRUNING PARAMETERS FOR EACH EXPERIMENT

Experiment	Pruning parameters
Brussels	n/a
GSL_B	n/a
GSL_{NB}	n/a
$MUSIC_{B,GLO}$	% retained = 50
$MUSIC_{NB,GLO}$	% retained = 50
$MUSIC_{B,LO}$	% retained = 42.5
$MUSIC_{NB,LO}$	% retained = 42.5
$AMUSES_{B,GLO}$	%retain = 0.05, %remove = 0.2, $thold_{low}$ = 0.00001 and $thold_{high}$ = 0.001
$AMUSES_{NB,GLO}$	%retain = 0.05, %remove = 0.2, $thold_{low}$ = 0.00001 and $thold_{high}$ = 0.001
$AMUSES_{B,LO}$	%retain = 0.05, %remove = 0.2, $thold_{low}$ = 0.00001 and $thold_{high}$ = 0.001
$AMUSES_{NB,LO}$	%retain = 0.05, %remove = 0.2, $thold_{low}$ = 0.00001 and $thold_{high}$ = 0.001
$M\text{-}CORE_B$	% retained = 50, l1-sam CORE = 0.01
$M\text{-}CORE_{NB}$	% retained = 50, l1-sam CORE = 0.01
M-CORE + ISD_B	% retained = 50, l1-sam CORE = 0.01, l1-sam ISD = 0.0001
M-CORE + ISD_{NB}	% retained = 50, l1-sam CORE = 0.01, l1-sam ISD = 0.0001

TABLE V

VALIDATION RESULTS FOR THE MAPPING EXPERIMENTS FOR EACH VIS CLASS

	Vegetation		Impervious		Soil	
Experiment	MAE	ME	MAE	ME	MAE	ME
Brussels	0.133	0.062	0.215	-0.156	0.149	0.089
GSL_B	0.114	0.071	0.236	-0.203	0.230	0.160
GSL_{NB}	0.118	0.068	0.245	-0.208	0.279	0.201
$MUSIC_{B,GLO}$	0.105	0.030	0.170	-0.140	0.163	0.138
$MUSIC_{NB,GLO}$	0.098	0.045	0.180	-0.148	0.173	0.123
$MUSIC_{B,LO}$	0.112	0.050	0.186	-0.140	0.160	0.102
$MUSIC_{NB,LO}$	0.120	0.052	0.180	-0.134	0.166	0.112
$AMUSES_{B,GLO}$	0.098	0.025	0.195	-0.168	0.163	0.140
$AMUSES_{NB,GLO}$	0.102	0.025	0.195	-0.167	0.190	0.155
$AMUSES_{B,LO}$	0.103	0.032	0.195	-0.135	0.165	0.108
$AMUSES_{NB,LO}$	0.103	0.045	0.190	-0.150	0.165	0.105
$M\text{-}CORE_B$	0.108	0.064	0.124	-0.078	0.080	0.030
$M\text{-}CORE_{NB}$	0.105	0.065	0.123	-0.075	0.080	0.033
M-CORE + ISD_B	0.096	0.050	0.112	-0.060	0.072	0.024
M-CORE + ISD_{NB}	0.102	0.064	0.122	-0.072	0.076	0.016

M-CORE with a distance-based pruning method (ISD) discarding redundant spectra from the library, a further reduction of the number of spectra in the pruned library could be achieved without substantially affecting mapping accuracies. Our research thus shows the merits of a hybrid, multistep approach to GSL pruning aimed at discarding image-irrelevant and redundant spectra, while limiting the impact of spectral confusion between classes on mapping accuracies. This idea can be conceptualized using the metaphor of the Swiss Cheese Model, where the three proposed steps in GSL pruning correspond with the different layers in the model (Fig. 8). Existing approaches for library pruning tend to focus on one or sometimes two of the above-mentioned steps (discarding irrelevant spectra and removing redundant spectra), but fail to sufficiently limit spectral confusion between spectra, a problem which in this study has been shown to significantly reduce GSL-based fraction mapping accuracies. Compared with MUSIC and AMUSES, M-CORE substantially improves class fraction accuracies, mainly because of the method's focus on selecting location-specific EM spectra that limit spectral confusion while still sufficiently capturing the spectral variability in the considered image subset. The results of our experiments also showed the potential of generic, multisite spectral libraries to support mapping applications. More in particular, our case study confirmed that, if multisite libraries are properly pruned, a GSL not including local spectra may produce class fraction estimates with accuracies comparable to those obtained with a GSL including both local and nonlocal spectra. This study thus highlights the benefits of library pruning and its importance for (semi-)automated GSL-based mapping applications, as well as the need for approaches addressing the three main concerns in library pruning. The setup of this proof-of-concept study was still quite specific in terms of site selection (Brussels), imagery used (Sentinel-2), GSL composition (mainly urban and European), and mapping method applied. More research is thus needed to study the merits of the proposed pruning approach and of GSL-based mapping methods in general, in a wider range of mapping setups. Moreover, the image segmentation into areas of dominant LC required by M-CORE should be made more data-driven, as it is currently based on user-defined segmentation rules. A more data-driven image segmentation would be particularly useful for mapping applications that consider more than three LC classes. In developing more generic workflows for GSL-based urban mapping, the use of M-CORE might also be coupled with the concept of urban ecoregions [19], where area-specific metadata such as climate, topography, dominant vegetation types, and morphological features of the built environment might be used as factors determining spectra selection.

V. CONCLUSION

In this paper, we demonstrated that applying a pruning technique that takes into account spectral redundancy, the relevance of spectra with respect to a particular site and spectral confusion can help in the development of more accurate and flexible remote sensing products.

APPENDIX A
PRUNING PARAMETERS

The pruning parameters for each experiment are summarized in Table IV.

APPENDIX B
VALIDATION RESULTS

The validation results are summarized in Table V.

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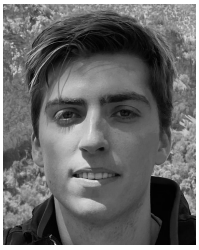
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